🏛️ GovFix Engine

Problem Understanding, Solution Proposal & System Architecture Report

VORTEXA 3.0 Hackathon
Domain 7: Government Systems & GovTech
Problem Statement: **G-02 GovFix**
Team: **OrbitX** | Oct 11, 2026

1. Problem Understanding

Public institutions—such as the **Pimpri Chinchwad Municipal Corporation (PCMC)**, **Pune Municipal Corporation (PMC)**, and large higher education campuses—face enormous volumes of citizen complaints and service requests (15,000+ monthly in Pune). These range from emergency infrastructure hazards (burst water mains, electrical cable exposures, road cave-ins) to routine maintenance.

Despite public intake portals (PCMC Sarathi, PMC Care, Samarth eGov), real-world grievance resolution breaks down due to **five structural failure modes**:

1. The "Limbo" Problem
Grievances enter "In Progress" and stall indefinitely with zero automated penalty, accountability, or progression forcing.

2. Ghost Resolutions
Contractors mark complaints "Resolved" without on-site photo evidence, physical inspection sign-offs, or citizen verification.

3. Silent SLA Breaches
Deadlines expire without notifying supervisors or municipal commissioners because standard databases lack proactive escalation daemons.

4. The Flaky AI Trap
Unconstrained LLM chatbots hallucinate state transitions, approve unauthorized closures, or drop records when given mutation rights.

> **Critical Failure: Standard databases (SQLite/MySQL) have mutable rows. If an officer delays a water contamination case, traditional logs cannot cryptographically prove when deadlines were missed or if records were tampered with.**

2. Proposed Solution: GovFix

GovFix is an institutional-grade, backend-enforced, finite-state workflow engine designed with one immutable governing rule:

> **"Deterministic Rules Govern Authority, State & Deadlines; AI Informs But Never Silently Overrides."**

GovFix replaces ad-hoc ticketing with four tightly integrated subsystems:

- **Deterministic 11-State Finite State Machine (FSM)**: Formally defines all legal paths. Illegal jumps (e.g. `SUBMITTED → RESOLVED`) are refused with machine-readable codes (`ERR_ILLEGAL_STATE_JUMP`) and `HTTP 409 Conflict`.
- **Backend-Enforced Role RBAC**: Every state mutation requires verified authority (`CITIZEN`, `WARD_OFFICER`, `FIELD_ENGINEER`, `SUPERVISING_COMMISSIONER`, `SYSTEM_DAEMON`). Unauthorized attempts return `HTTP 403 Forbidden`.
- **Dynamic Multi-Tier SLA Escalation Engine**: Automatically assigns deadlines based on urgency (12h for Hazards, 24h for High, 48h for Standard, 72h for Routine). When deadlines breach, background daemons escalate to Tier-1 (Ward Officer) and Tier-2 (Municipal Commissioner).
- **Tamper-Evident SHA-256 Cryptographic Audit Ledger**: Every state change, escalation, and *refused attempt* is chained into an immutable SHA-256 block sequence, proving 100% mathematical integrity.

- **Virtual Time-Travel Simulation**: Interactive controls (`+12h` , `+24h` , `+48h`) allow evaluators to test multi-day escalation workflows in seconds.

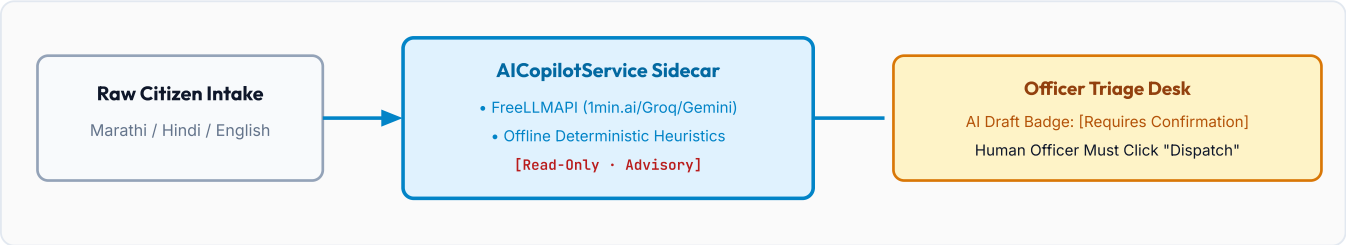
3. Intended Users & Role Boundaries

GovFix enforces strict, backend-guarded boundaries across five institutional actors:

User Role	Permitted Actions	Prohibited Actions (Strictly Refused)
Citizen / Requester	• Submit grievance (Marathi/Hindi/English) • Track live status & SLA countdown • Confirm resolution with satisfaction score • Reopen within 48h if unsatisfied	• Cannot triage or assign officers • Cannot unilaterally mark tickets Resolved • Cannot alter SLA deadlines
Ward Officer / Supervisor	• Review side-by-side AI triage recommendations • Override category, ward, or priority • Reject invalid spam with logged justification • Dispatch work orders to Field Engineers • Conduct physical inspection sign-offs	• Cannot skip engineering repair stage • Cannot resolve complaints without inspection • Cannot delete audit records
Field Engineer / Staff	• Acknowledge assigned work orders • Transition ticket to `IN_PROGRESS` • Upload photographic evidence & completion notes • Request `INSPECTION_PENDING`	• Cannot approve own work as final `RESOLVED` • Cannot reassign tickets to other staff • Cannot modify SLA timers
Supervising Commissioner	• City-wide SLA compliance oversight • Review active Tier-1 and Tier-2 escalations • Issue direct administrative orders on overdue cases • Reallocate emergency ward resources	• Cannot bypass deterministic state flow • Cannot erase audit ledger entries
System SLA Daemon	• Continuous background time evaluation • Automated transitions to `ESCALATED_L1` / `L2` • Emits real-time SSE alerts	• Cannot close or resolve tickets

4. Constrained AI Integration (FreeLLMAPI + Heuristic Fallback)

GovFix strictly complies with statutory safety requirements: **AI is an advisory sidecar with zero state-mutation authority.**



- **Multilingual Grievance Extraction**: Ingests raw Marathi (*"रावेत चौक रस्त्यावर मोठा खड्डा पडला आहे"*), Hindi, or English and proposes `{ category, urgency, ward, draft_response }`.
- **Proximity Duplicate Detection**: Identifies complaints within 200m in the same ward to eliminate redundant work orders.
- **100% Offline Resilience**: If external APIs or internet are severed, an embedded keyword and regex heuristics engine ensures core operations continue with zero downtime.

5. Differentiation From Existing Solutions

Capability / Criterion	Samarth eGov (ERP)	ManageEngine ServiceDesk	Cloud ITSM (Freshservice)	GovFix (Team OrbitX)
Workflow Core	Form submission list	Hardcoded statuses	Proprietary cloud rule engine	**Deterministic 11-State FSM**
Refusal Handling	Generic validation failure	Generic UI error	Generic UI error	**Exact HTTP 409 & 403 Refusal Codes**
Audit Ledger Integrity	Mutable SQL rows	Application database logs	Vendor cloud logs	**Tamper-Evident SHA-256 Hash Chain**
Refused Actions Logged	❌ Ignored	❌ Ignored	❌ Ignored	✅ **Immutable record with Actor & Role**
Multi-Tier Escalation	Basic email reminder	Notification trigger	Email notification	**Automated Tier-1 & Tier-2 Daemon Transitions**
Evaluator Time-Travel	❌ None	❌ None	❌ None	✅ **Virtual Clock (+12h, +24h, +48h)**
Deployment & Cost	Heavy ERP deployment	High enterprise license	Expensive monthly SaaS	**Zero Paid APIs · 100% Local Run MVP**

6. System Architecture & Workflow Diagram

The architecture cleanly separates user channels, authorization enforcement, workflow logic, audit immutability, and advisory AI:

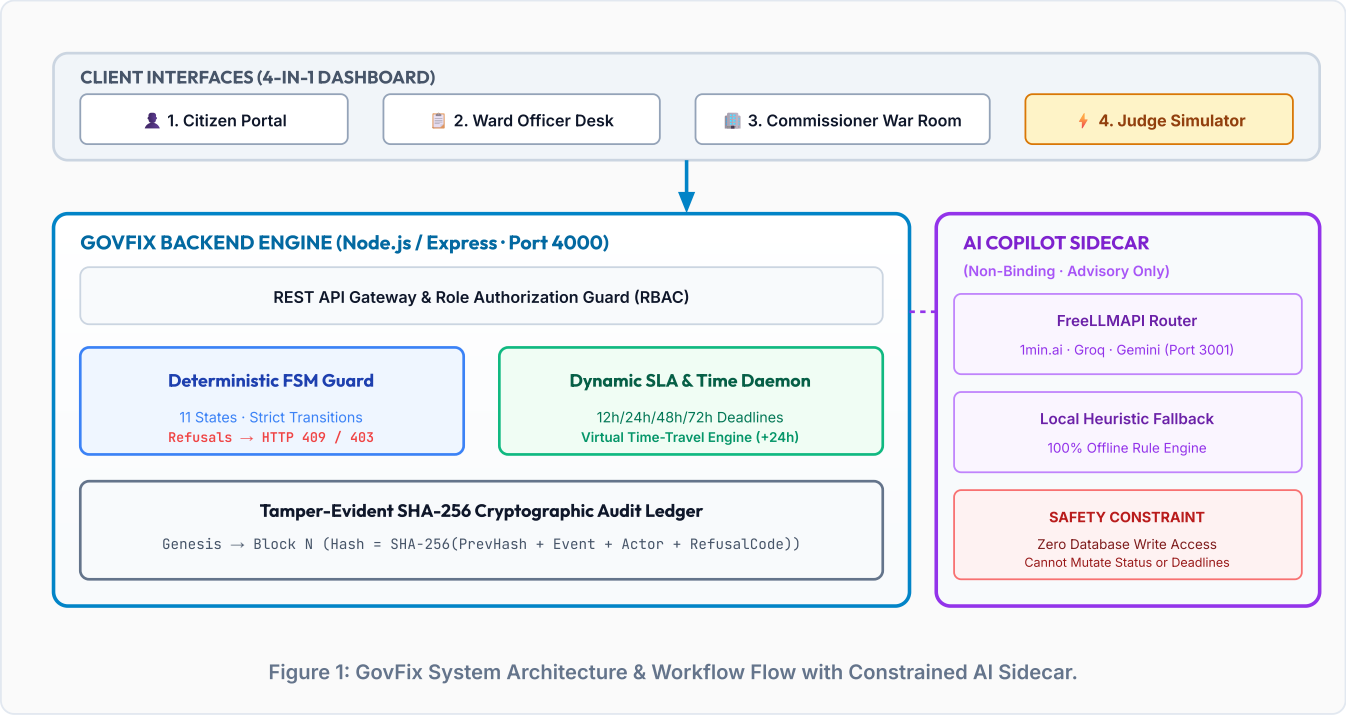


Figure 1: GovFix System Architecture & Workflow Flow with Constrained AI Sidecar.

7. Verification & Synthetic Testing Results

GovFix has been verified through an automated end-to-end integration test suite (`npm test`) mapping directly to the official benchmark criteria:

#	Test Name	Observed Result	Status
1	API Server & Health	REST endpoint responds with active grievance list	PASS
2	Multilingual Citizen Submission	Marathi text ingested; category & 24h SLA assigned	PASS
3	Deterministic Refusal (Illegal Jump)	SUBMITTED → RESOLVED rejected with code `ERR_ILLEGAL_STATE_JUMP` [409]	PASS
4	RBAC Role Authorization Refusal	Citizen attempting officer triage rejected with `ERR_ROLE_UNAUTHORIZED` [403]	PASS
5	Ward Officer Triage	Valid transition SUBMITTED → TRIAGED approved	PASS
6	Field Work Order Dispatch	Valid transition TRIAGED → ASSIGNED with Engineer ID	PASS
7	Virtual Time-Travel Escalation	+24h time-advance triggers automated Tier-1 escalation	PASS
8	Cryptographic Chain Verification	100% SHA-256 forward hashes mathematically valid	PASS

Summary: Team OrbitX has delivered a 100% locally runnable, rigorously tested, tamper-evident workflow engine ready for live evaluation at VORTEXA 3.0.